Steps for Task 1: Understanding Data Types

For activity one I have sampled retail sales data from Kaggle (2023). The data, which is in CSV form, has nine columns as listed below:

· Transaction ID: Each transaction receives a unique number that helps us identify it.

· Date: Date of Event: Object (Transaction Timing).

· Customer ID: Record ID identifies every person who makes purchases.

· Gender: The records contain data about the purchaser under the label Gender with values Male or Female.

· Age: Integer (Customer's age).

· Product Category: Product Type represents a distinct product grouping.

· Quantity: The last purchase figure as a unique numeric number.

· Price per Unit: A basic integer shows how much one-unit costs.

· Total Amount: The transaction shows its full cost as an integer.

Ordinal Data

Ordinal data points follow a natural ranking system for their categories, yet their measured distance remains unknown (Juan Luis Suárez, García and Herrera, 2021). Our newly added Age Group entry demonstrates ordinal data characteristics. These age categories follow a standard progression with clear ranking positions from Child to Young Adult to Adult to Senior. The categories "Child" and "Young Adult" stand lower than "Adult," with all ages following in this ranking system. While we can determine these categories order in a clear sequence, the actual age distance between them stayed unknown. The ordering of these categories makes the variable ready for turning labels into numbers while keeping the rank relationships intact, yet data analysts should not view these numbers as standard measurements.

Categorical Data

In this dataset, we can identify two types of categorical variables: nominal and ordinal. The different categories in these variables have no selected positions when ranked. The dataset includes two categories named "Genre" (such as Male and Female) and "Product Category" (Electronic and Clothing), which stand as isolated labels without sub-levels. The placement of items exists in ordinal data attributes because these values exhibit rank relationships (Chen and Anderson, 2024). The dataset now contains the Age Group variable, which uses the expected life stages Child, Young Adult, Adult, and Senior. When working with nominal data, one-hot encoding avoids suggested ordering while label-encoding remains the preferred method for ordinal data types.

Encoding Categorical Variables

Nominal variables show different groups that rank equally (such as "Gender" and "Product Category"). Machine learning models need one-hot encoding to process nominal variables. The method generates one binary column for each category present in a nominal variable. For example, the "Gender" column will be transformed into two new columns: The data receives a 1 in the "Gender_Male" or "Gender_Female" column aligned with its category plus zeros in all other entries. This process stops from adding unnatural order and fits well with machine learning algorithms.

Preparing Data for Machine Learning Models

First, the data is prepared by summarizing the whole dataset and getting a view of the first few rows of each column of data.

```
RangeIndex: 1000 entries, 0 to 999
Data columns (total 9 columns):
 #   Column            Non-Null Count  Dtype
---  ------            --------------  -----
 0   Transaction ID    1000 non-null   int64
 1   Date              1000 non-null   object
 2   Customer ID       1000 non-null   object
 3   Gender            1000 non-null   object
 4   Age               1000 non-null   int64
 5   Product Category  1000 non-null   object
 6   Quantity          1000 non-null   int64
 7   Price per Unit    1000 non-null   int64
 8   Total Amount      1000 non-null   int64
dtypes: int64(5), object(4)
memory usage: 70.4+ KB
(   Transaction ID        Date Customer ID  Gender  Age Product Category  \
 0               1  2023-11-24     CUST001    Male   34           Beauty
 1               2  2023-02-27     CUST002  Female   26         Clothing
 2               3  2023-01-13     CUST003    Male   50      Electronics
 3               4  2023-05-21     CUST004    Male   37         Clothing
 4               5  2023-05-06     CUST005    Male   30           Beauty

    Quantity  Price per Unit  Total Amount
 0         3              50           150
 1         2             500          1000
 2         1              30            30
 3         1             500           500
 4         2              50           100  ,
 None)
```

Figure 1: Machine Learning Model preparations

For the Machine learning models the updated processed dataset now integrates original and newly encoded column information to support machine learning modelling. Here's a breakdown of the changes and additions:

Original Numerical Columns

These are the initial unprocessed numeric fields from the original dataset, such as:

- Transaction ID: Every transaction needs its individual recognition code.
- Age: Customer age in years.
- Quantity: How many products did a customer buy at one time.
- Price per Unit: Price of each item.
- Total Amount: The total transaction amount equals how many items times their unit purchase price.

Encoded Columns for Categorical Variables

New columns have been created to transform categorical data into a format usable by machine learning models:

- Age Group Encoded: Our encoding shows the ordered Age Group values starting with Child set to 0 then moving through Young Adult (1), Adult (2), and ending with Senior at 3. These rankings stay intact as they were before.
- Gender_Male: The gender values have been encoded in a binary column showing either 1 for Male or 0 for Female.
- Product Category_Clothing, Product Category_Electronics, etc.: Infomal One-Hot systems transform "Product Category" values into separate columns representing distinct category items. For example: When a product matches "Clothing" it receives a value of 1 in the Product Category_Clothing field, and within the Electronics product group Product Category_Electronics shows a 1 value.

Output From the Python Code

Transaction ID	Date	Customer ID	Age	Quantity	Price per Unit	Total Amount	Age Group Encoded	Gender_Male	Product Category_Clothing	Product Category_Electronics
1	2023-11-24	CUST001	34	3	50	150	3	True	False	False
2	2023-02-27	CUST002	26	2	500	1000	3	False	True	False
3	2023-01-13	CUST003	50	1	30	30	0	True	False	True
4	2023-05-21	CUST004	37	1	500	500	0	True	True	False
5	2023-05-06	CUST005	30	2	50	100	3	True	False	False

Figure 2: Encoded Columns Output

1. Regression Analysis

For my dataset because the "Total Amount" column shows continuous data it works well as a target for regression. The column shows us the exact price of every transaction.

Python snip code

```
# Categorize Age into groups (Ordinal Data)
dataset['Age Group'] = pd.cut(dataset['Age'], bins=[0, 18, 35, 60, 100],
```

```
                    labels=['Child', 'Young Adult', 'Adult', 'Senior'])
# Identify categorical data
nominal_columns = ['Gender', 'Product Category']
ordinal_columns = ['Age Group']
```

Output

```
Available columns: Index(['Transaction ID', 'Date', 'Customer ID', 'Age', 'Quantity',
       'Price per Unit', 'Total Amount', 'Age Group Encoded', 'Gender_Male',
       'Product Category_Clothing', 'Product Category_Electronics'],
      dtype='object')
R² Score: 0.86
Root Mean Squared Error (RMSE): 205.85
Mean Absolute Error (MAE): 174.47
```

Interpretation of Results

After running the code, the regression model performed as follows:

R^2 Score: 0.86: The model uses 86% of all available data to predict Total Amount accurately. An R^2 score of 0.86 shows strong connections between our input data metrics (quantity, price per unit) and the output metric we are trying to predict. Our model shows strong results by understanding the connection between variables.

Root Mean Squared Error (RMSE): The Root Mean Square Error shows how far predictions miss their target when reported in the same numbers as Total Amount (Chicco, Warrens and Jurman, 2021). The model produces forecasts with an average difference of $205.85 from actual results. In a business application, this error size would be either minor or major depending on normal transaction values (small when most processes handle amounts in the thousands).

Mean Absolute Error (MAE): MAE shows how far apart actual results stand from their predicted values with an average standard unit measurement (Chicco, Warrens and Jurman, 2021). An MAE signal of 174.47 shows the model consistently misses real values by 174.47-dollar units.

Since MAE shows lower sensitivity to large errors than RMSE, these results show that prediction errors remain reasonably scattered.

2. Classification Analysis

We perform gender (male or female) prediction using features that include age, quantity, price per unit, and encoded product categories. We can use a binary classification problem since the target variable (Gender) has two classes: Male and Female. For the data above, we will implement two classification models: logistic regression and random forest classifier.

Model: Logistic Regression

✅ Accuracy: 0.46

✅ Precision: 0.45

✅ Recall: 0.46

✅ F1 Score: 0.45

🔹 Confusion Matrix:

[[69 84]
 [79 68]]

🔹 Classification Report:

	precision	recall	f1-score	support
0	0.47	0.45	0.46	153
1	0.45	0.46	0.45	147
accuracy			0.46	300
macro avg	0.46	0.46	0.46	300
weighted avg	0.46	0.46	0.46	300

The predictive performance of logistic regression above proved unsatisfactory because the model achieved an accuracy of 46%. In this scenario the predictive model achieved an accuracy level of 46% through correct classification of test data samples. The model

demonstrates inadequate capability in class differentiation as indicated by its failure to achieve predictable results and recall efficiency since both metrics measure 45% and 46%, respectively. The model's 0.45 F1 score demonstrates precision-recall equilibrium, yet it reveals its limited capability to address this classification challenge. A nearly equivalent distribution between false positive (84) and false negative cases (79) appears in the confusion matrix because the model fails to distinguish both types correctly.

Model: Random Forest

✅ Accuracy: 0.47

✅ Precision: 0.46

✅ Recall: 0.48

The Random Forest prediction model operated at a 47% accuracy rate and demonstrated a precision score of 46% and a recall rate of 48%. Performance metrics from the random forest model slightly surpass logistic regression results but show overall inadequate model performance. Weak patterns together with insufficient predictive capability of present features likely explain why the models display only minimal variations in predictive accuracy (El-Shafiey et al., 2022). Future work should address performance deficiencies by performing feature engineering as well as dataset resampling and structural evaluation.

3. *K-Means Clustering*

Elbow Method for Optimal K

An illustration of the Elbow Method presents a visualization that enables users to select K-Means clustering's ideal cluster number (Darshan Anil Jethva et al., 2023). Within-Cluster Sum of Squares (WCSS), also known as inertia, operates as the measure of cluster compactness and appears on the y-axis. As K (number of clusters) changes, it appears along the x-axis. The

location of the "elbow" in the graph demonstrates the optimal number of clusters (K). The best cluster numbers fall between K=4 and K=5 based on the elbow in the chart.

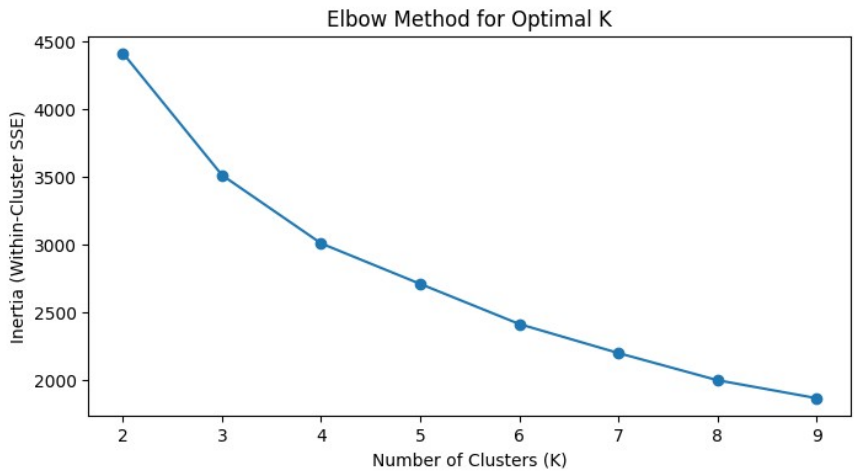

Figure 3: K-means Elbow Diagram

Customer Segmentation Using K-Means

The graph below shows customer segmentation achieved through K-Means clustering. The data received two-dimensional representation through dimensional reduction using Principal Component Analysis (PCA) to yield PC1 and PC2 components. A different colour coding scheme illustrates separate clusters which reveal distinct customer segments from their features. The separated clusters demonstrate that K-Means properly placed similar customers into their respective groups (Darshan Anil Jethva et al., 2023). Clustering analysis delivers potent value for business segmentation purposes. Business utilizes cluster grouping strategies in customer behaviour analysis to generate data-driven strategic decisions that deliver improved satisfaction and profitability for its users.

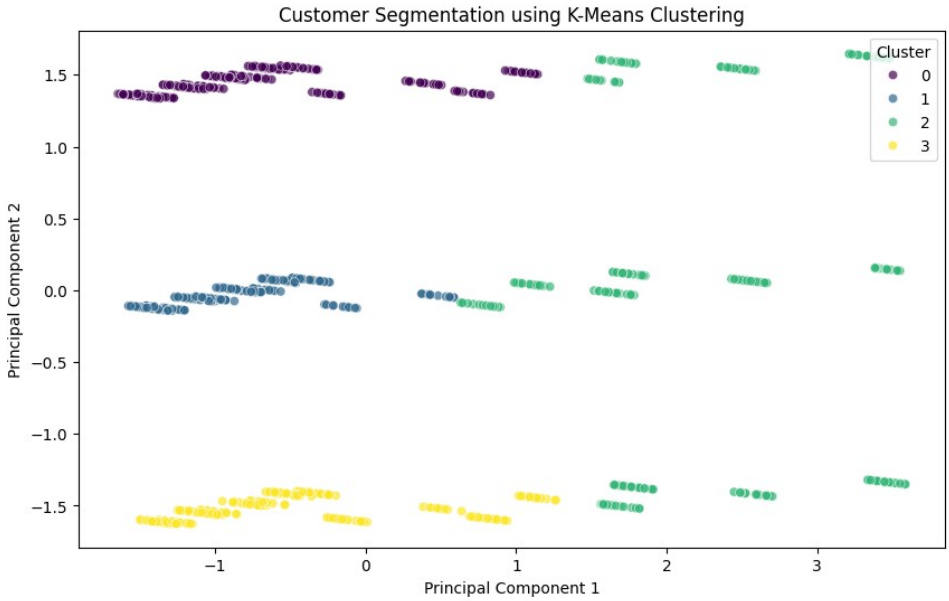

Figure 3: K-means Cluster Diagram

Task 3: Project Life Cycle – End-to-End

Business Problem

Customer churn prediction for subscription-based service products represents the business problem to analyze and solve. The cancellation of customer subscriptions, which results in revenue loss, is known as churn. Maintaining current clients proves less expensive than seeking out fresh subscribers for a business. The company can predict future customer departures so it can execute proactive retention measures, which include tailored offers and reducing costs for remaining subscribers.

How Analytics Solves It

A predictive model enables evaluation of customers into two groups: those "likely to churn" and those "not likely to churn." Through the development of predictive models, the

business can prioritize its retention efforts toward high-risk customers, which both minimizes customer loss and generates higher profits.

Data Preparation, Cleaning, and Splitting

We will use the provided website to source data that contains subscription services; the source we found at UC Irvine ML (2021) was for a dataset gathered randomly over 12 months from an Iranian telecom company database. We would use Python code to clean and prepare it before deciding on an appropriate model to use for prediction. Since the data was huge and we needed more accurate results, the data was split into 70% and 30% for the training and testing, respectively.

Modelling

The Random Forest Classifier will be utilised to predict customer churn, utilizing its ability to address different classification scenarios when dealing with unbalanced datasets despite nonlinear relationships and multivariable interactions. This model, according to Cota (2023), supplies insight into which features most powerfully persuade customers to leave so that we can pinpoint the essential factors causing customer churn.

From the Python code, the following output from the random forest prediction was identified:

```
Random Forest Performance:
Accuracy: 0.9439153439153439
Confusion Matrix:
 [[771 ,12]
 [ 41, 121]]
Classification Report:
              precision    recall  f1-score   support
...
    accuracy                           0.94       945
   macro avg       0.93      0.87      0.89       945
weighted avg       0.94      0.94      0.94       945
```

Random Forest shows superior outcomes to Logistic Regression through all evaluation metrics especially when identifying churn occurrences. That model delivered superior results in terms of precision and recall and F1-score alongside improved recall capabilities for class 1 when compared to logistic regression.

Bibliography

Chen, D. and Anderson, C. (2024). Categorical Data Analysis. *arXiv (Cornell University)*, 5(9). doi:https://doi.org/10.48550/arxiv.2409.02942.

Chicco, D., Warrens, M.J. and Jurman, G. (2021). The Coefficient of Determination R-squared Is More Informative than SMAPE, MAE, MAPE, MSE and RMSE in Regression Analysis Evaluation. *PeerJ Computer Science*, [online] 7(5), p.e623. doi:https://doi.org/10.7717/peerj-cs.623.

Cota, X. (2023). Addressing data imbalance in customer churn prediction: A novel approach for telecommunications companies. *Handle.net*, [online] 14(6). doi:http://hdl.handle.net/10071/30986.

Darshan Anil Jethva, Siya Milind Khamkar, Anish Anand Pachchhapur and Kulkarni, S. (2023). Customer Segmentation Analysis using K-means Algorithm with Elbow method and Dendrogram. *Customer Segmentation Analysis using K-means Algorithm with Elbow method and Dendrogram*, 12(13). doi:https://doi.org/10.1109/icast59062.2023.10454952.

El-Shafiey, M.G., Hagag, A., El-Dahshan, E.-S.A. and Ismail, M.A. (2022). A hybrid GA and PSO optimized approach for heart-disease prediction based on random forest. *Multimedia Tools and Applications*, 81(13), pp.18155–18179. doi:https://doi.org/10.1007/s11042-022-12425-x.

Juan Luis Suárez, García, S. and Herrera, F. (2021). Ordinal regression with explainable distance metric learning based on ordered sequences. *Machine Learning*, 110(10), pp.2729–2762. doi:https://doi.org/10.1007/s10994-021-06010-w.

UC Irvine ML (2021). *UCI Machine Learning Repository*. [online] archive.ics.uci.edu. Available at: https://archive.ics.uci.edu/dataset/563/iranian+churn+dataset.